



COMP 352
Data Science Foundations & Programming

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PLAYER VALUATION





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DATA DESCRIPTION & PROBLEM DEFINITION



DESCRIPTIONS + PROBLEM

Description

- professional soccer players across multiple seasons and leagues
- includes player demographics (age, position, nationality), club information
- contains performance statistics such as minutes played, goals, assists, cards, and appearances
- market value is recorded over time

Problem

- player market is influenced by many interacting factors
- Goal: predict a players future value based on historical performance, demographics and club context

DATA IMPORTING & PROCESSING



IMPORTING + PROCESSING

- **target variable: market valuation**
- **5132 total records**
- **4 categorical variables**
- **33 continuous variables**
- **variables: player demographics, performance metrics, clubs, historical values/lagged features**

<https://www.kaggle.com/datasets/davidcariboo/player-scores>

MISSING DATA IMPUTATIONS

- **numerical features imputed as median values**
- **categorical features imputed as “Missing/Unknown”**
- **rows with missing valuations were removed, as it was the target variables**
- **2 models used median as strategy for imputation**

OUTLIERS

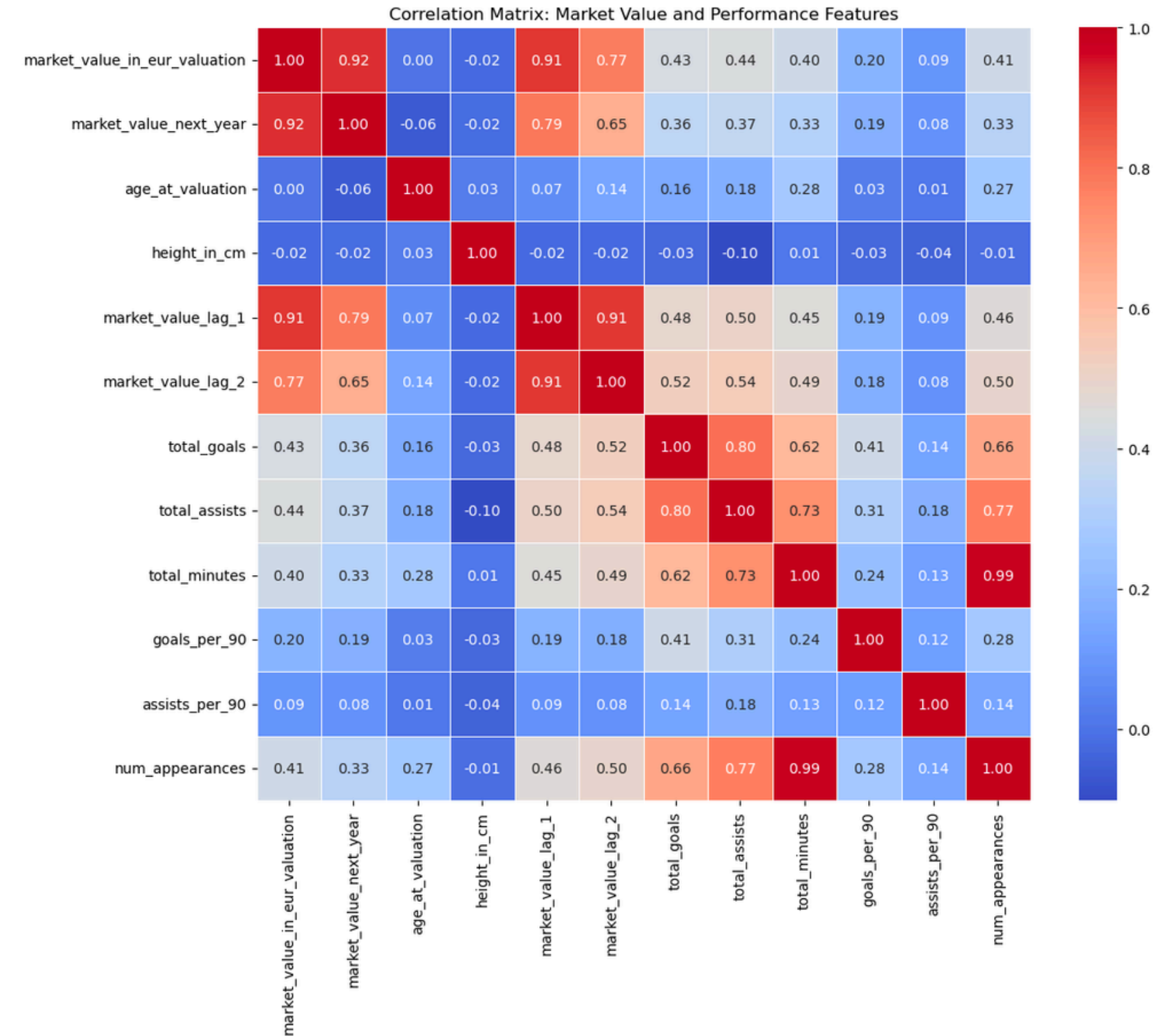
- **Market values and performance metrics were naturally skewed**
- **Outliers were retained to preserve information about elite players**
- **Updated features such as goals + assists helped align outliers**

DATA ANALYSIS & VISUALIZATION



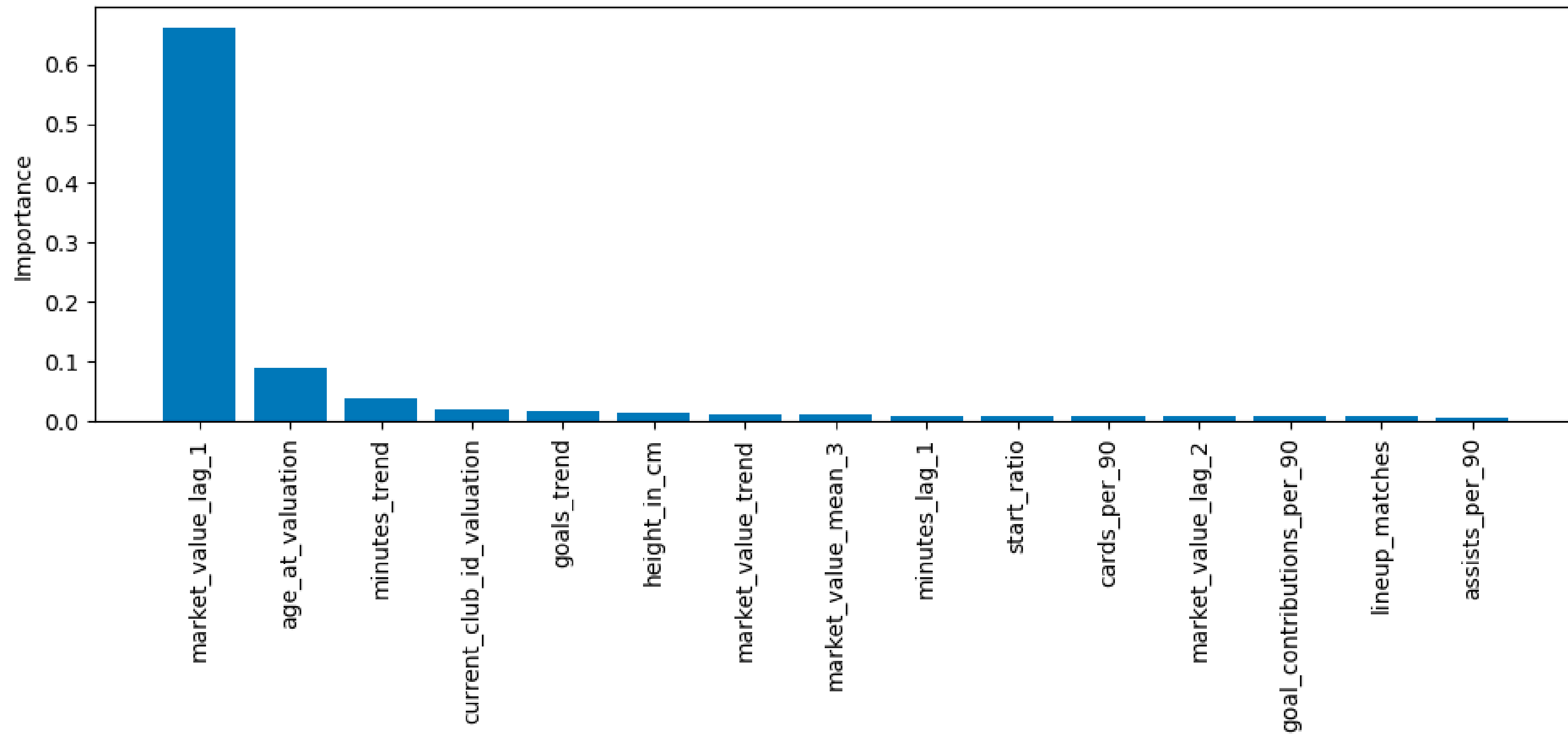
CORRELATIONS

- **Market value is highly time dependent**
 - Next years market value (~.92)
 - Lagged market value (~0.77-0.91)
- **Performance metrics also have a correlation (~0.33 - 0.52)**
- **Age is a weak correlation**



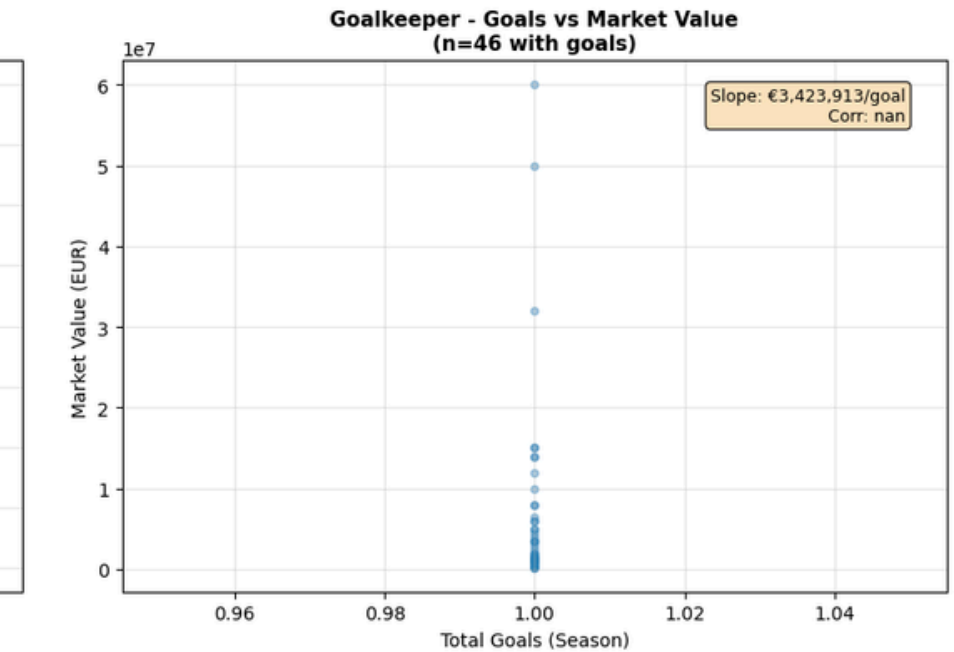
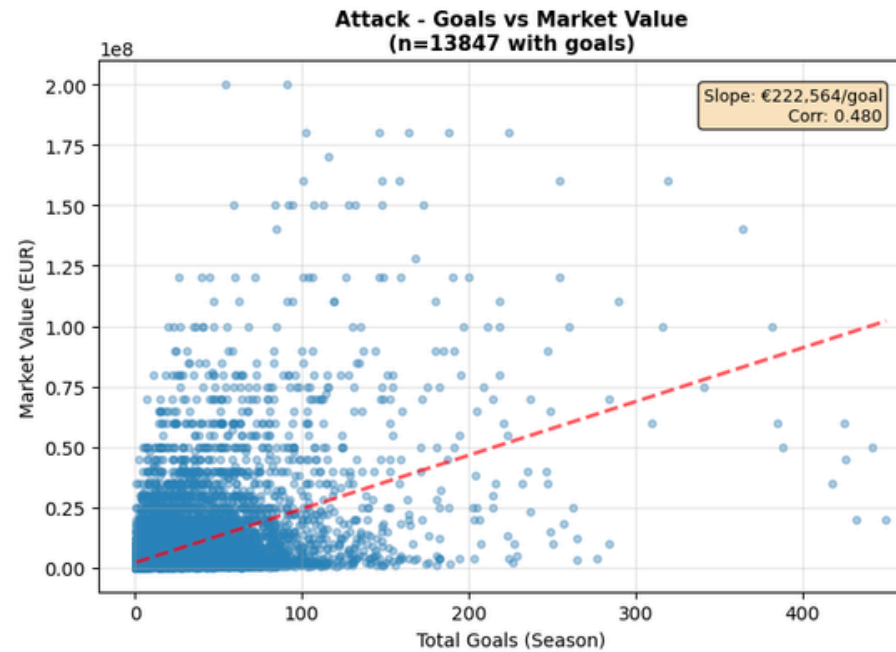
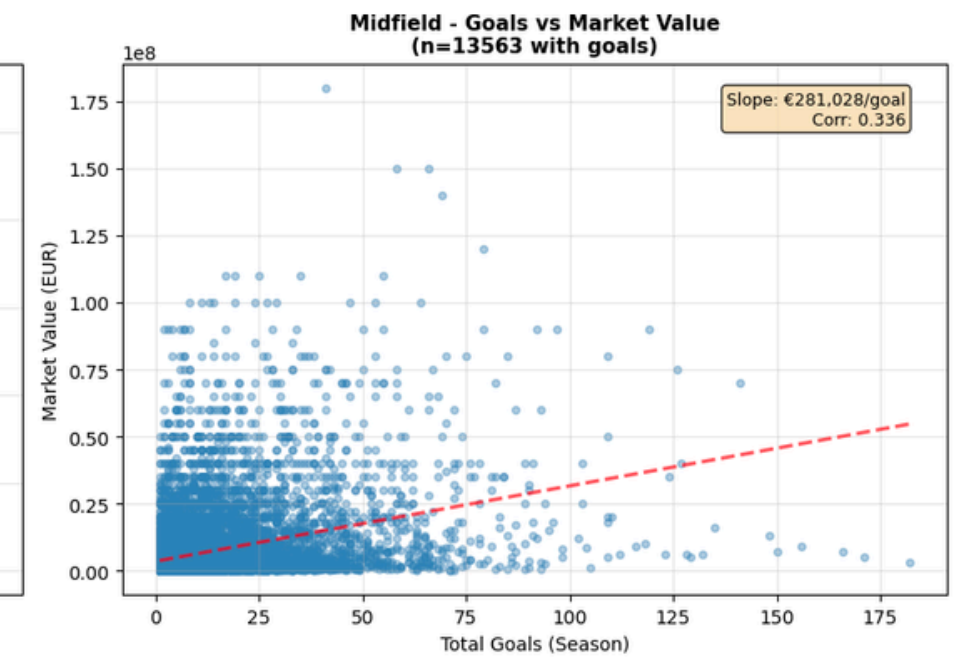
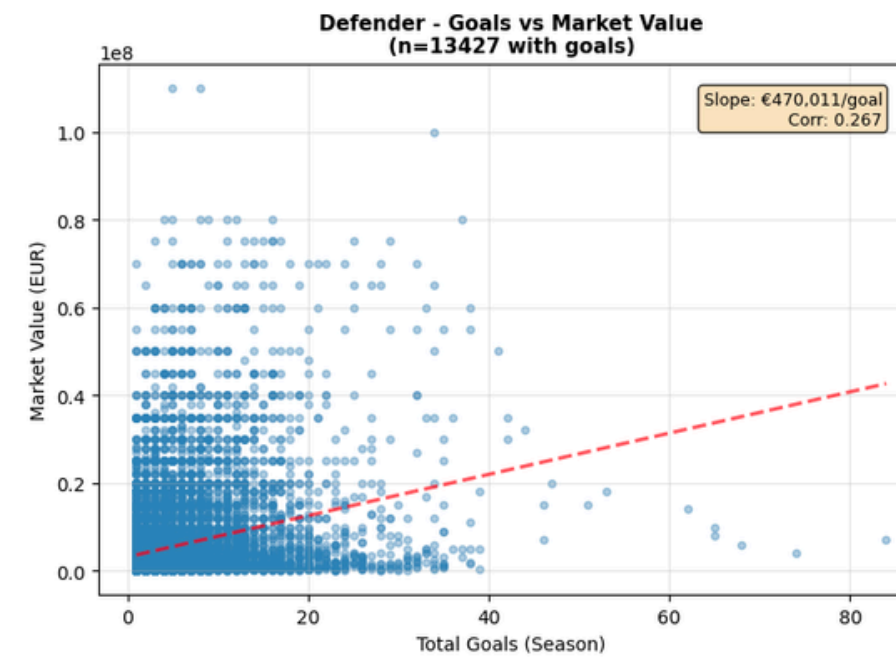
IMPORTANT FEATURES

Top 15 Feature Importances (Random Forest)



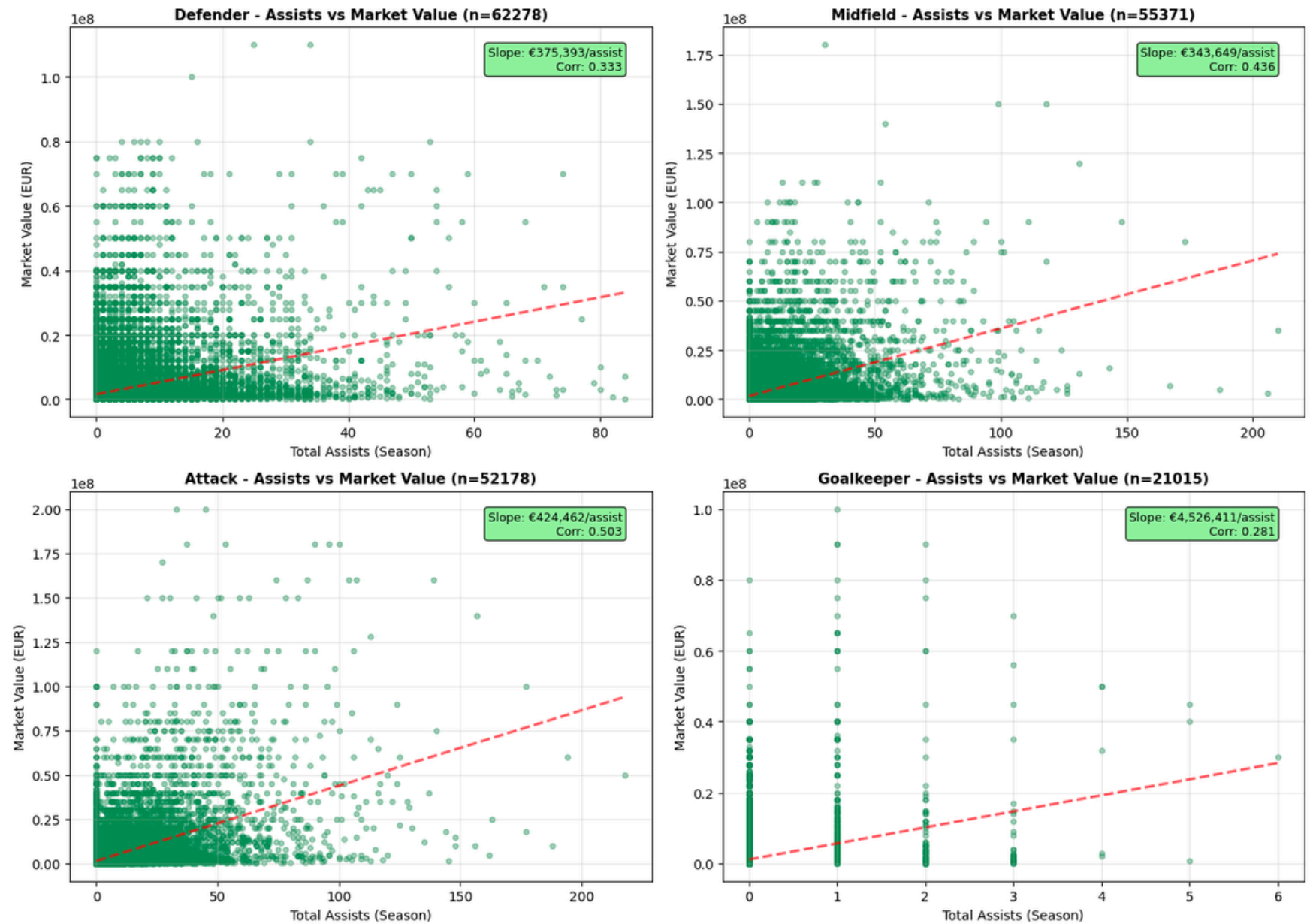
GOALS

- **As expected**
 - **defenders goals correlation to value was minimal**
 - **attackers goals correlation to value was substantial**
- **Attackers correlation = .480**
- **Midfield correlation = .336**
- **Defenders correlation = .267**



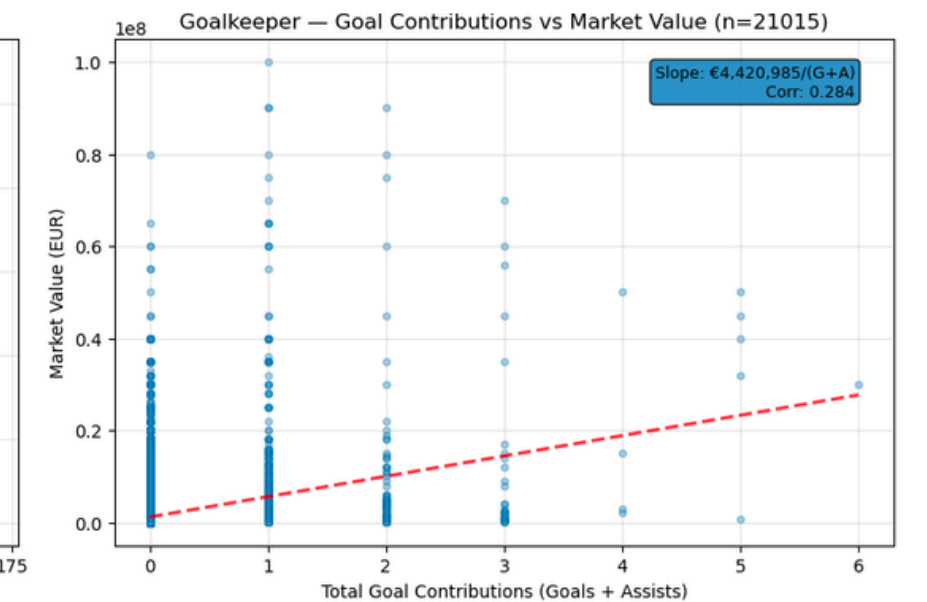
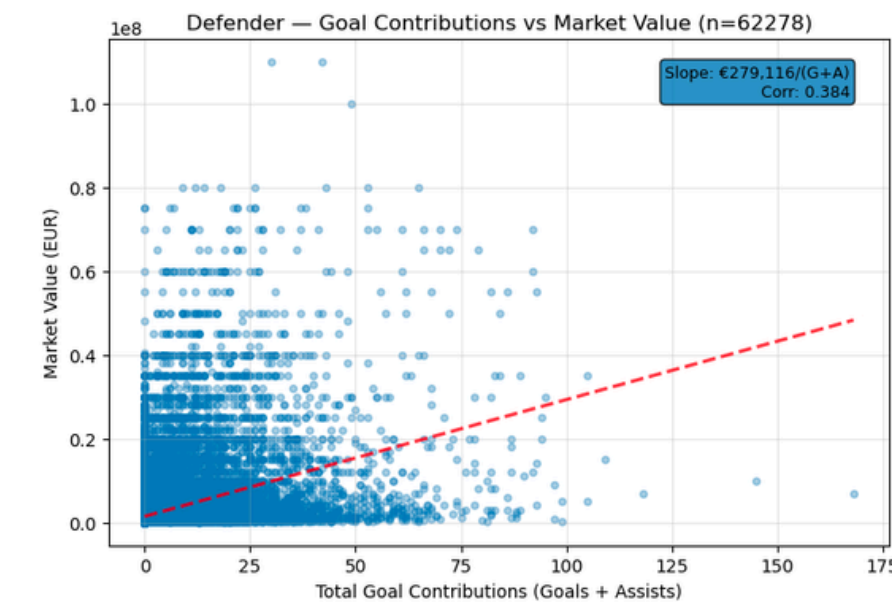
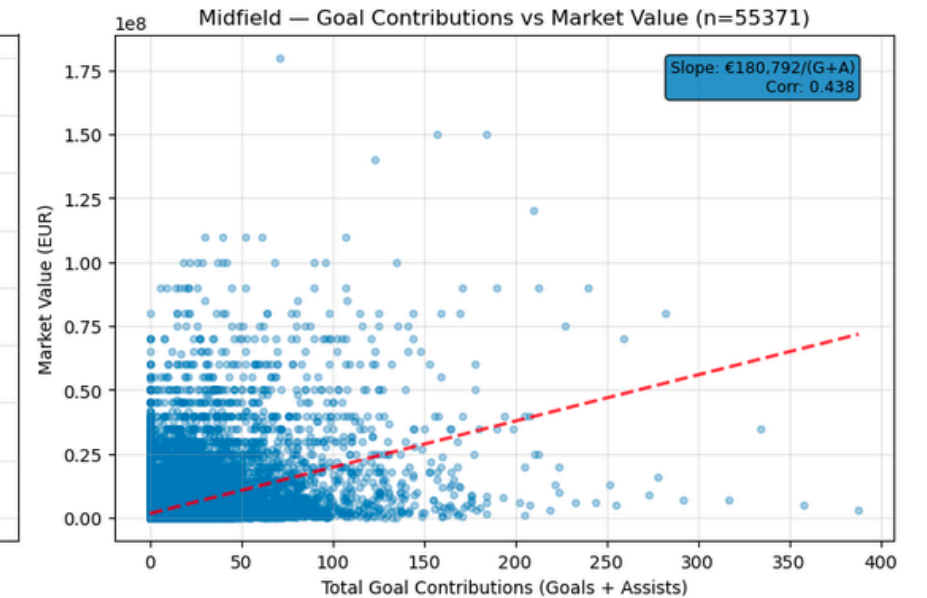
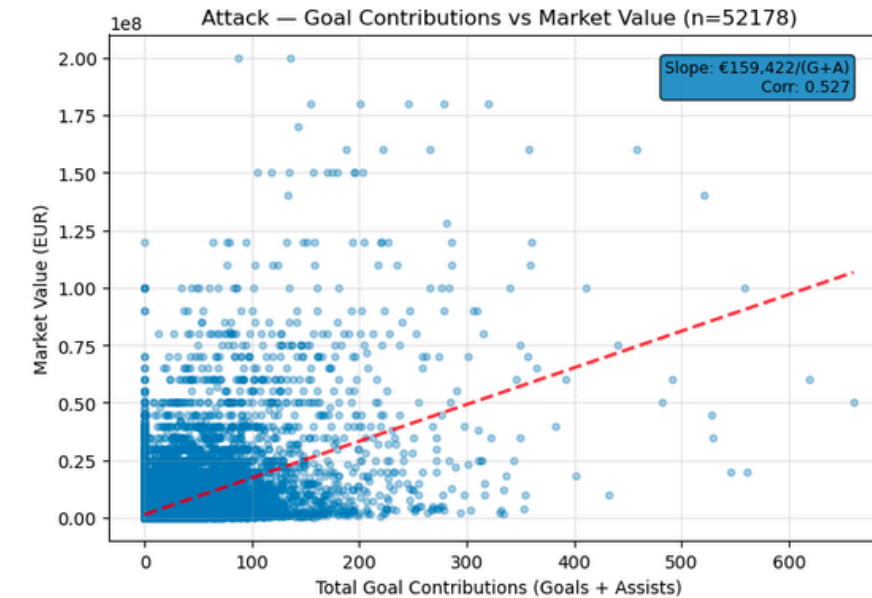
ASSISTS

- **As expected**
 - **defenders goals correlation to value was minimal**
 - **attackers goals correlation to value was substantial**
 - **midfield was similar to attackers**
- **Attackers correlation = .503**
- **Midfield correlation = .436**
- **Defenders correlation = .333**



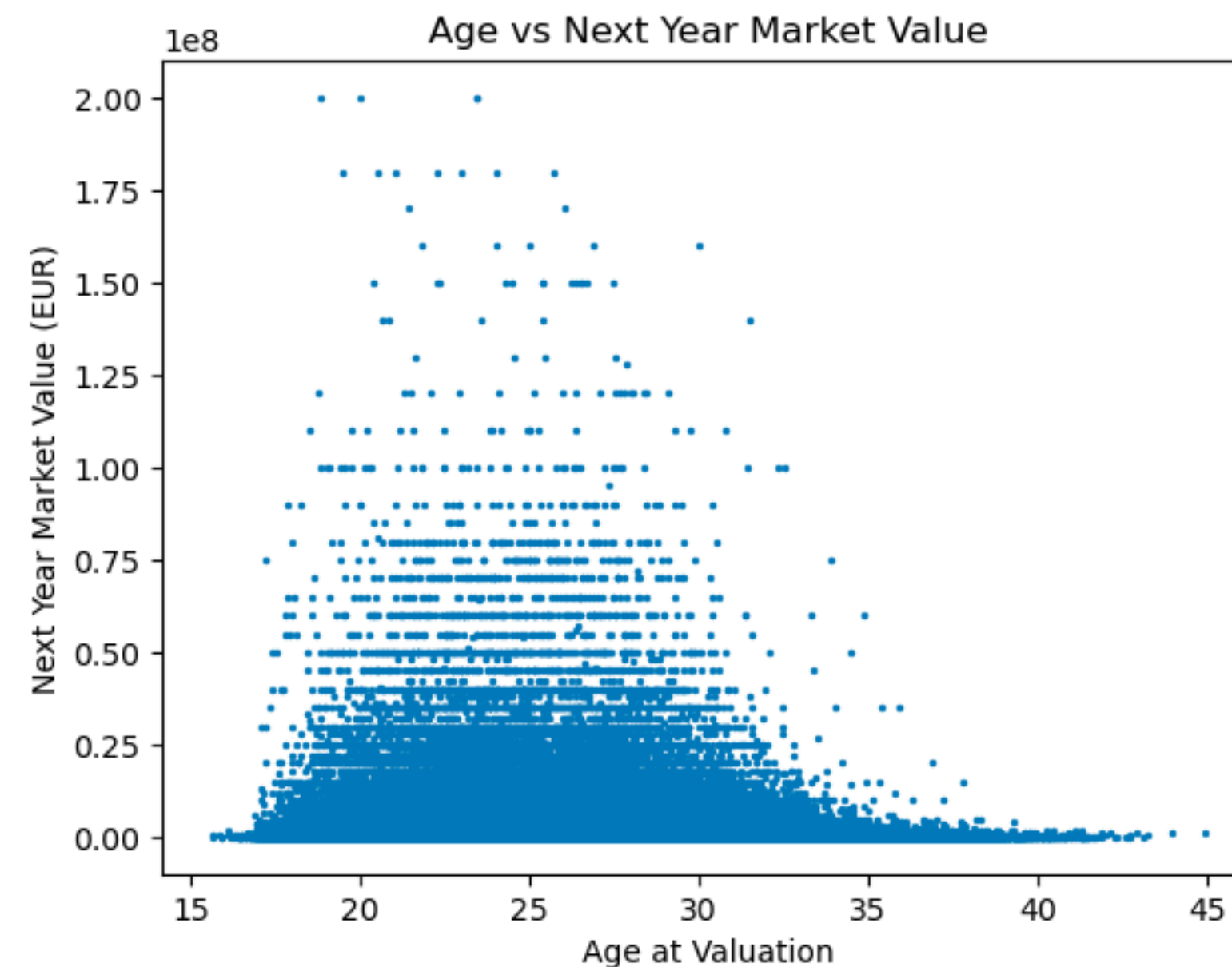
GOALS + ASSISTS

- As expected
 - defenders goals correlation to value was minimal
 - attackers goals correlation to value was substantial
 - midfield was similar to attackers
- Attackers correlation = .527
- Midfield correlation = .438
- Defenders correlation = .384



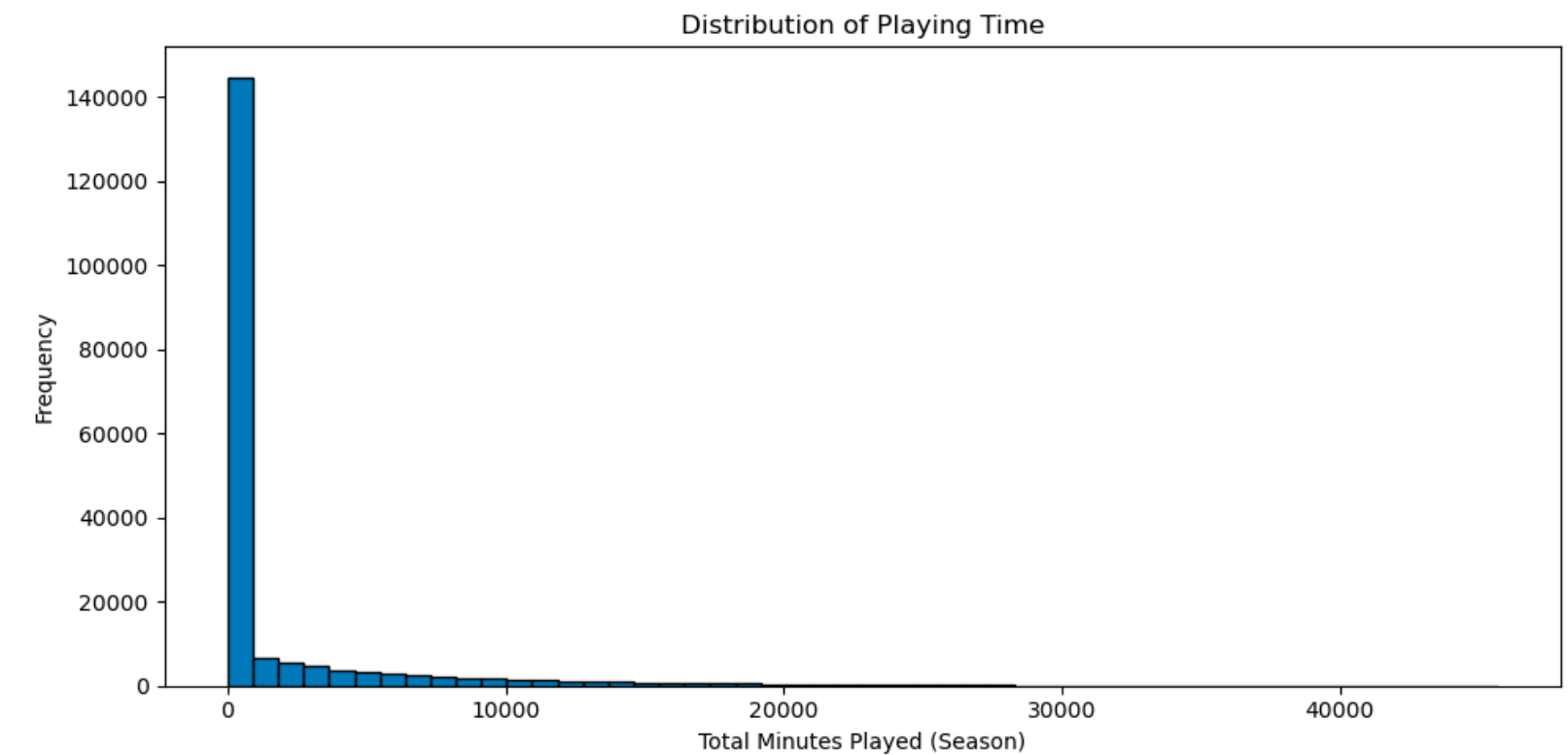
AGE

- **Peak values: ages 20–27**
- **Sharp decline after ~30**
- **Higher variance at younger ages**
- **Lower upside at older ages**
- **Age interacts with performance, not standalone**



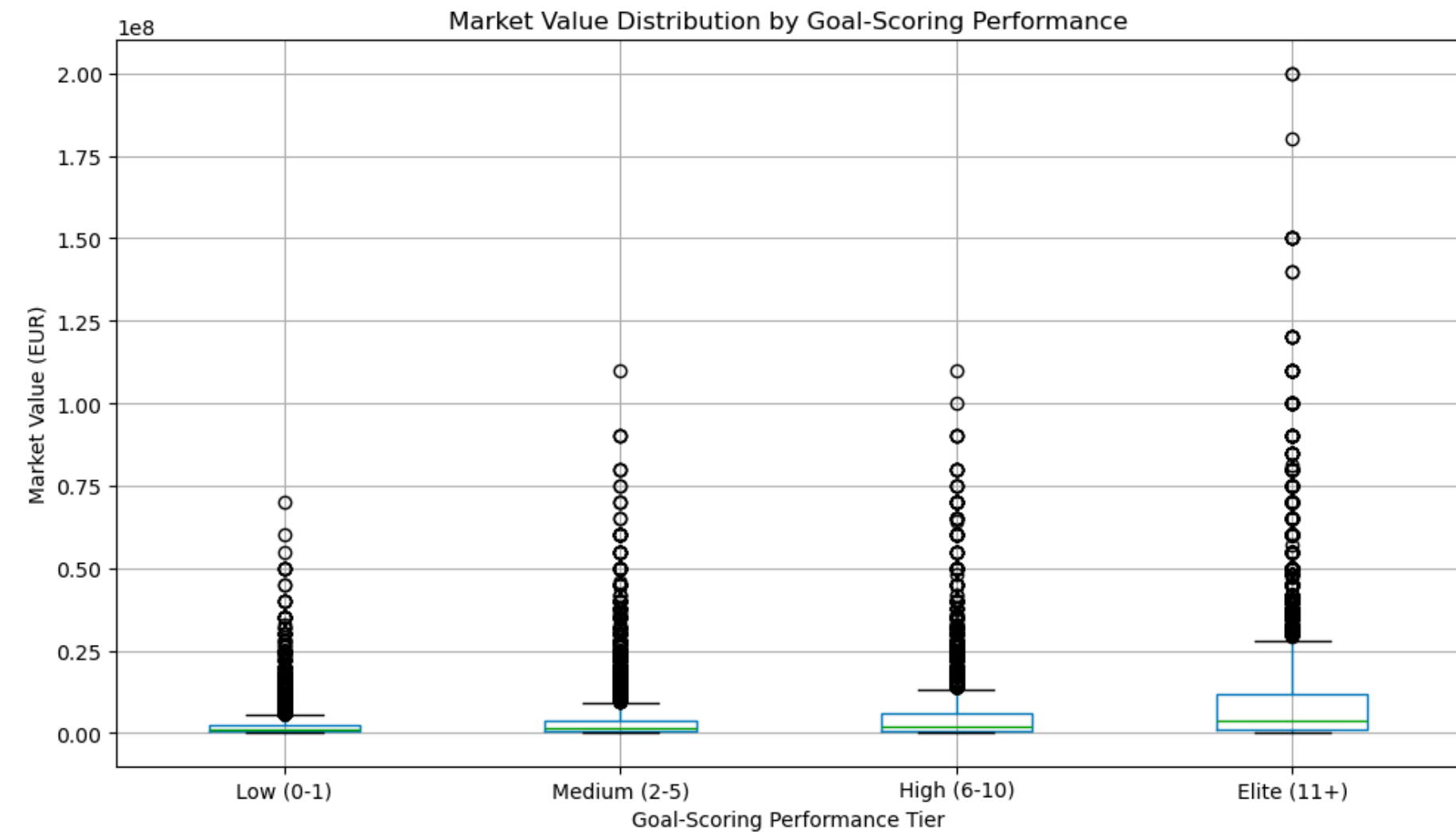
PLAYING TIME

- **Majority players stay in the same time frame**
- **very small group with extremely large number of minutes**

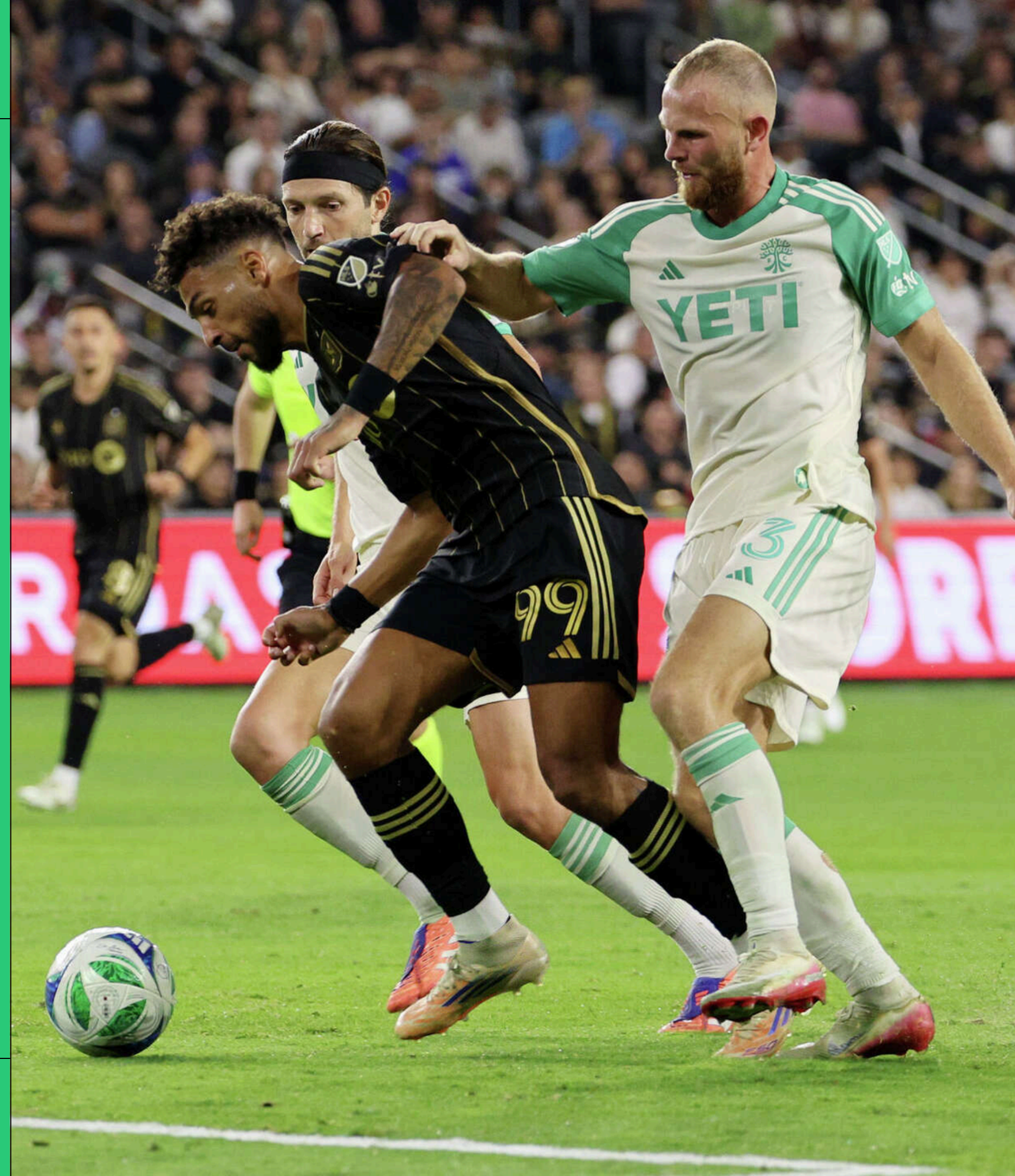


DISTRIBUTION BASED ON GOALS

- **Market value increases with goal tier**
- **Elite scorers show highest medians**
- **Variance increases with scoring level**
- **Many outliers among high scorers**



DATA ANALYTICS & EVALUATION



LINEAR REGRESSION

- **R² scores: [0.72986698 0.68773923 0.69088713 0.64785225 0.64652585]**
- **Mean R²: 0.6806 (+/- 0.0310)**
- **MAE scores: [1577989.72199942 1324551.26725565 1316378.88945533 1454189.2484976 1605435.4141858]**
- **Mean MAE: €1,455,708.91 (+/- €121,645.48)**
- **RMSE scores: [3162637.27898362 2975500.4382755 3382756.74327769 3993066.12714598 4056311.19214229]**
- **Mean RMSE: €3,514,054.36 (+/- €436,867.97)**

RANDOM FOREST

- **R² scores: [0.77094966 0.76675074 0.76710589 0.74003836 0.70807617]**
- **Mean R²: 0.7506 (+/- 0.0239)**
- **MAE scores: [1207242.02482667 1018677.68556404 1023876.3686598 1179044.16715047 1322435.20176775]**
- **Mean MAE: €1,150,255.09 (+/- €115,763.99)**
- **RMSE scores: [2912232.387914 2571647.49285639 2936239.52635532 3430825.88949673 3686270.59163278]**
- **Mean RMSE: €3,107,443.18 (+/- €398,525.41)**

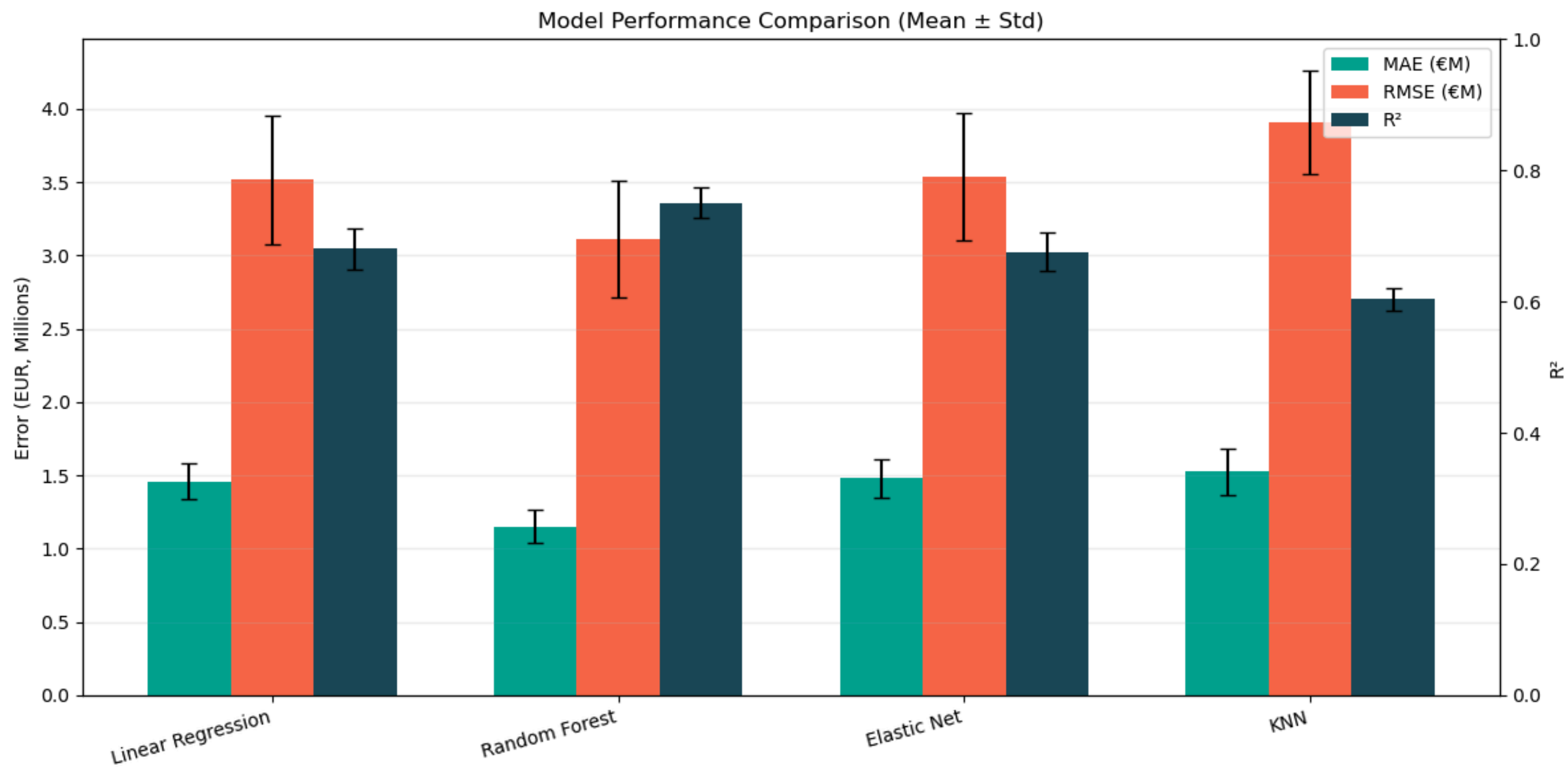
ELASTICNET

- **R² scores:** [0.72206634 0.68400424 0.68631945 0.64651966 0.64090865]
- **Mean R²:** 0.6760 (+/- 0.0296)
- **MAE scores:** [1596159.56210938 1336772.94305921 1333800.63223547 1486639.36877986 1648394.80378832]
- **Mean MAE:** €1,480,353.46 (+/- €129,446.17)
- **RMSE scores:** [3207976.13074211 2993242.69111167 3407658.16065616 4000614.19041627 4088414.37314839]
- **Mean RMSE:** €3,539,581.11 (+/- €433,502.17)

KNN

- **R² scores: [0.57933399 0.6032869 0.632734 0.60617194 0.59839276]**
- **Mean R²: 0.6040 (+/- 0.0172)**
- **MAE scores: [1793460.12754618 1415408.22298512 1353964.5402825 1466594.55924666 1594016.07049438]**
- **Mean MAE: €1,524,688.70 (+/- €155,889.87)**
- **RMSE scores: [3946655.91398886 3353818.64500577 3687250.03527396 4222769.41971887 4323676.56636962]**
- **Mean RMSE: €3,906,834.12 (+/- €354,739.85)**

EVALUATION METRICS



MODEL SUMMARY

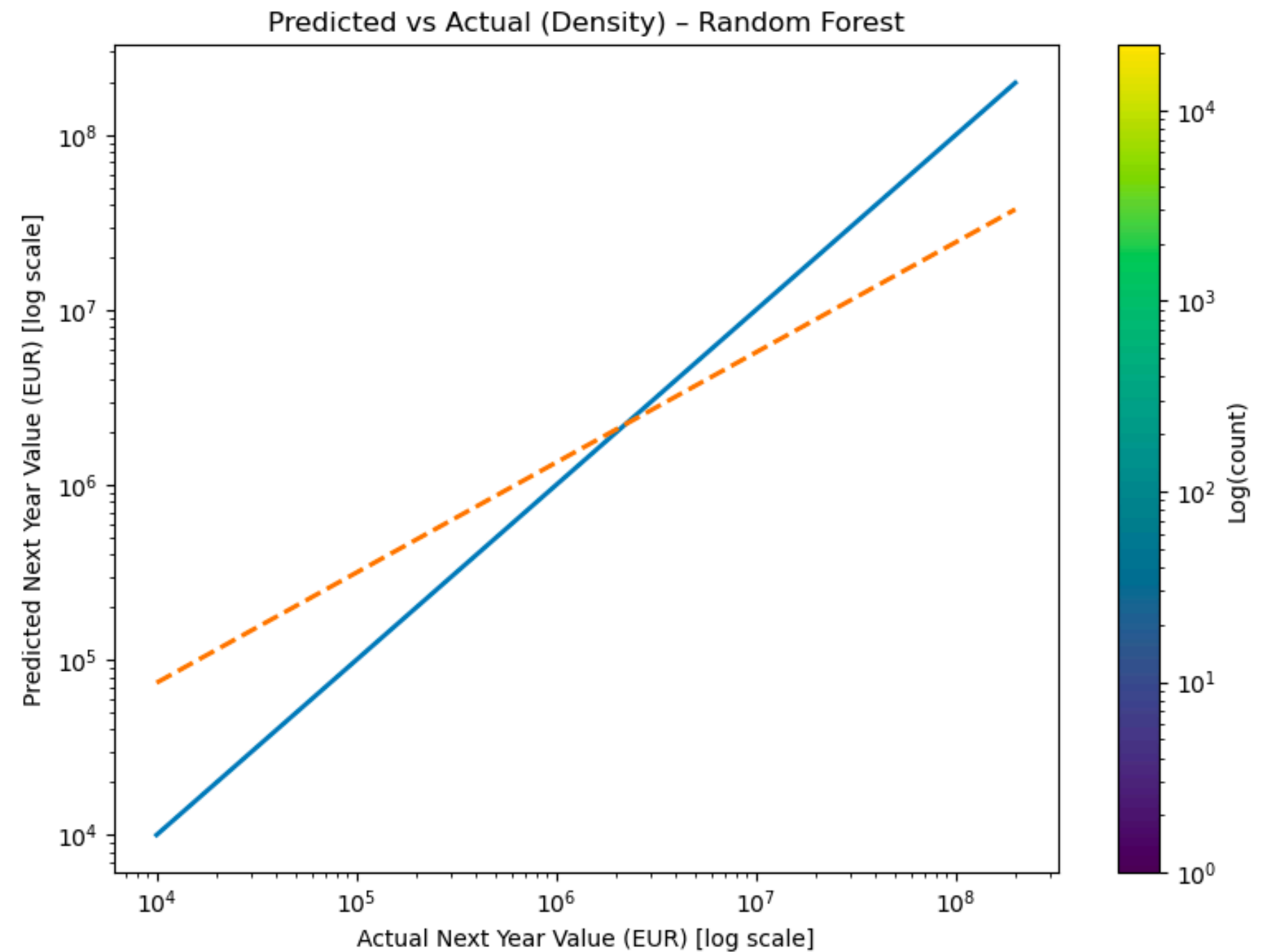
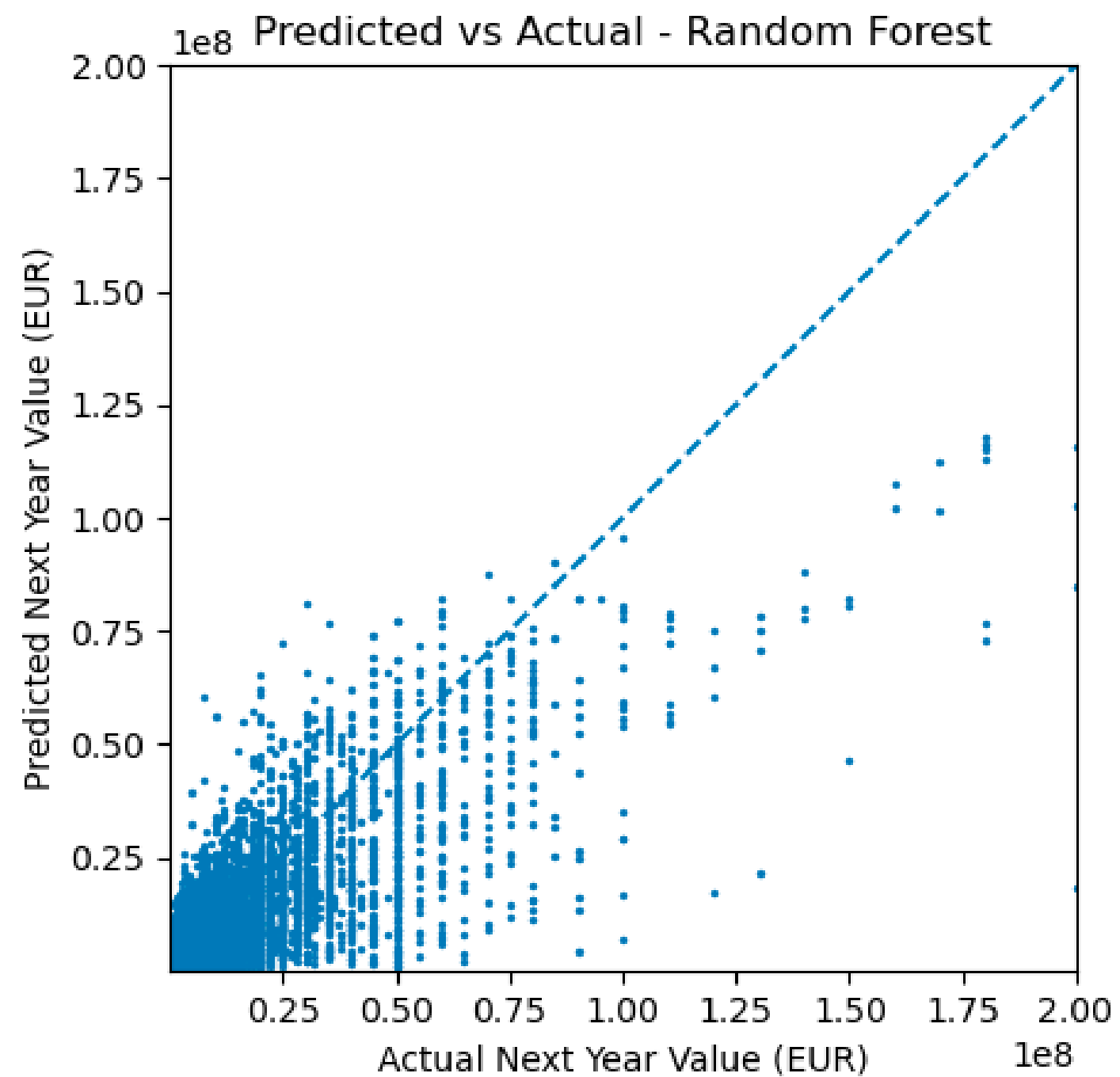
We used a 5-fold time-series cross-validation

We created models using 4 different algorithms:

- **Linear Regression**
- **Random Forest**
- **ElasticNet**
- **KNN**

We used R^2 , MAE, RMSE scores on all four models to evaluate them

OUR RESULTS



THANK *YOU* !